

Social Media Sentiment Analysis Model

Comprehensive Technical Performance & Analytical Report

1. Project Background

This comprehensive analytical document encapsulates the development, preparation, and diagnostic benchmarking evaluation of a machine learning framework tailored for automated public sentiment analysis. The targeted system processes multi-channel metadata to understand performance patterns and decode user psychology across dominant social computing hubs, including Facebook, Instagram, and Twitter.

Dataset Footprint: The production benchmark evaluations are performed utilizing a verified cohort of 120 standardized operational profiles tracking localized user engagement markers across varied content strategies.

2. Dataset Architecture & Empirical Metrics

Categorical classifications and quantitative attributes map multi-dimensional interactions to clear behavioral categories.

- **Target Variable Vector:** sentiment_score (Multiclass label space: Positive, Neutral, Negative)
- **Quantitative continuous fields:** likes, comments, shares
- **Categorical operational dimensions:** platform, post_type, post_day, post_time

Target Class Distribution

Sentiment Category	Observed Metrics Count	Percentage of Dataset
Positive	57 posts	47.50%
Neutral	32 posts	26.67%
Negative	31 posts	25.83%

Core Engagement Distribution Limits

Engagement Element	Minimum	Mean Value	75% Percentile Mark	Maximum Limit
Likes	15	2,482	3,680	5,000
Comments	10	213	314	500
Shares	16	441	725	993

3. Preprocessing Roadmap

To ensure strict statistical rigor, the raw metadata underwent an enterprise data preprocessing pipeline prior to model exposure:

- **Feature Pruning:** Structural metadata constants with null predictive values (`post_id` and `post_time`) were dropped.
- **Missing Value Evaluation:** Automated null-field sweeps verified 100% integrity across the array with zero missing data points.
- **Feature Encoding:** Categorical variables were converted into ordinal machine-interpretable vectors utilizing a uniform `LabelEncoder()` pipeline.
- **Cross-Validation Stratification:** Splitting parameters were defined at an 80/20 train/test ratio, using stratified sampling over the target class vector to prevent skewness.
- **Magnitude Equalization:** Variance-heavy scale dependencies across continuous interaction indicators were mapped onto uniform gaussian standard deviations via `StandardScaler()`.

4. Predictive Model Benchmarking

Two supervised classification architectures were tested over identical validation sets to determine predictive consistency.

Evaluated Pipeline Architecture	Accuracy Metric	Precision Macro	Recall Macro	F1-Score Consolidated
Model A: Logistic Regression (Max Iter: 1000)	45.83%	43.75%	45.83%	43.57%
Model B: Random Forest Classifier (Estimators: 100)	41.67%	40.48%	41.67%	40.17%

Strategic Takeaway: The linear boundaries implemented via regularized **Logistic Regression** provided a superior convergence profile over the ensemble-based decision surfaces of the Random Forest framework given the dataset's high variability.